INTERNSHIP PROJECT REPORT

ON

ONLINE QUIZ MANAGEMENT SYSTEM

A Web-Based Platform for Creating, Conducting
and Evaluating Online Quizzes

**Submitted in partial fulfillment of the requirements
for the Internship Program**

Submitted By: **Shivam singh**

Roll no : 25scs1003000974

Section: 2cse38

Under the Guidance of: intern, cse / bluestocks

CERTIFICATE

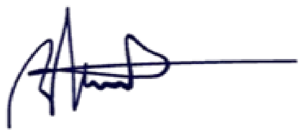
This is to certify that the project report entitled “Online Quiz Management System” submitted by shivam singh student of Your College iilm university, is a bona fide record of the work carried out by him/her under my supervision and guidance during the internship period.

The project work embodied in this report has not been submitted earlier for the award of any degree or diploma to the best of my knowledge.

Date: 1 july 2026

Place: 2 july 2026

Signature of the Guide
Ganesh karkale



DECLARATION

I hereby declare that the internship project report entitled “Online Quiz Management System” submitted by me is an authentic work carried out by me under the guidance of [Guide Name]. The work presented in this report is original and has not been submitted earlier for any academic degree or diploma.

I further declare that all the information and data presented in this report are true to the best of my knowledge and belief.

Date: 30 august 26

Place: grtear noida

Shivam singh

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me throughout the successful completion of this internship project on Online Quiz Management System.

First and foremost, I am deeply grateful to my project guide, [Guide Name], for his/her continuous guidance, valuable suggestions, and constant encouragement. His/her expertise and insights helped me overcome technical challenges and shape this project into a complete solution.

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Shivam singh

ABSTRACT

In the modern educational landscape, traditional paper-based examinations are increasingly being replaced by digital solutions that offer greater efficiency, accuracy, and accessibility. This project presents the design and development of an Online Quiz Management System — a comprehensive web-based platform that enables educators to create, manage, and conduct quizzes while allowing students to attempt them remotely with real-time evaluation.

The system supports multiple user roles including Administrator, Teacher, and Student, each with tailored functionalities. Teachers can create quizzes with multiple-choice, true/false, and subjective questions, set timers, configure negative marking, and publish assessments. Students can register, attempt timed quizzes, and receive instant results. The system automatically evaluates objective questions and generates detailed performance reports.

The application is built using modern web technologies with a focus on security (role-based access, password hashing), scalability, and responsive design. Key features include auto-save of answers, browser focus detection for basic anti-cheating measures, and comprehensive analytics dashboards.

This report documents the complete software development lifecycle of the project — from requirements analysis and system design (including architecture, ER diagrams, and use cases) to implementation, testing, and future enhancements. The developed system successfully addresses the limitations of traditional examination methods and provides a robust foundation for digital assessment in educational institutions.

Keywords: Online Quiz System, Web Application, E-Learning, Automated Evaluation, Role-Based Access Control, Educational Technology

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1. INTRODUCTION

1.1 Background

Education systems worldwide are undergoing a rapid digital transformation. The COVID-19 pandemic accelerated the adoption of online learning platforms, making remote assessment not just a convenience but a necessity. Traditional examination methods involving paper, pen, and manual evaluation are time-consuming, prone to human error, and environmentally unsustainable.

An Online Quiz Management System addresses these challenges by providing a centralized digital platform where quizzes can be created, distributed, attempted, and evaluated with minimal human intervention. Such systems are widely used in schools, colleges, coaching institutes, corporate training programs, and competitive examination platforms.

1.2 Problem Statement

Despite the availability of commercial learning management systems, many educational institutions still rely on paper-based tests or rudimentary digital tools that lack features such as automatic evaluation, role-based security, detailed analytics, and mobile responsiveness. The key problems identified are:

- Manual evaluation is slow and error-prone.
- Difficulty in conducting simultaneous large-scale assessments.
- Lack of instant feedback for students.
- No centralized question bank or reusable content.
- Limited performance tracking and reporting capabilities.
- Security and integrity concerns during remote exams.

This project aims to develop a practical, secure, and user-friendly Online Quiz Management System that overcomes these limitations.

1.3 Objectives

The primary objectives of this project are:

1. To design and develop a web-based system for creating and managing online quizzes.
2. To implement automatic evaluation for objective-type questions with instant result generation.
3. To provide role-based access control for Admin, Teacher, and Student users.
4. To support timed quizzes with auto-submit functionality and basic anti-cheating measures.
5. To generate detailed performance reports and analytics for continuous improvement.
6. To ensure the system is responsive, secure, and scalable for institutional use.

1.4 Scope of the Project

The scope of this project includes the development of a complete web application covering user authentication, quiz creation and management, question bank, timed examination, automatic scoring, result display, and basic reporting. The system is designed for educational institutions and can be extended for corporate training. Advanced features such as AI proctoring, video monitoring, and adaptive testing are considered as future enhancements and are outside the current scope.

2. LITERATURE SURVEY & EXISTING SYSTEMS

Several commercial and open-source platforms provide online assessment capabilities. Popular tools include Google Forms, Microsoft Forms, Moodle Quiz module, Kahoot, Quizizz, and specialized platforms like ExamSoft and ProctorU. While these tools offer robust features, many institutions require a lightweight, customizable, and cost-effective solution that can be hosted on their own servers and integrated with existing systems.

Research in e-assessment highlights the importance of automatic scoring, item analysis, and immediate feedback for improving learning outcomes. Studies also emphasize the need for secure authentication, randomized question presentation, and lockdown browser features to maintain examination integrity.

Existing systems often suffer from one or more of the following limitations: high licensing costs, limited customization, dependency on third-party cloud services, or insufficient analytics. The proposed Online Quiz Management System aims to fill this gap by offering an open, modular, and institution-friendly solution.

Comparison with Existing Tools

Feature	Google Forms	Moodle	Proposed System
Automatic Evaluation	Yes (limited)	Yes	Yes
Role-Based Access	Basic	Advanced	Advanced
Timer & Auto-submit	Limited	Yes	Yes
Detailed Analytics	Basic	Yes	Yes
Self-Hosted Option	No	Yes	Yes
Cost	Free	Free / Paid	Low / Free
Customization	Limited	High	High

3. SYSTEM ANALYSIS

3.1 Functional Requirements

The system must support the following functional requirements:

Administrator

- Register and manage user accounts (Teachers and Students).
- Create and manage quiz categories/subjects.
- View system-wide statistics and reports.
- Configure system settings and security policies.

Teacher / Faculty

- Create, edit, and delete quizzes.
- Add questions of multiple types (MCQ, True/False, Subjective).
- Set quiz duration, total marks, and negative marking rules.
- Publish or unpublish quizzes for specific student groups.
- View individual and aggregate student performance reports.
- Export results in PDF or Excel format.

Student

- Register and log in securely.
- View list of available and upcoming quizzes.
- Attempt quizzes within the allotted time.
- Receive instant results after submission.
- View attempt history and performance trends.

3.2 Non-Functional Requirements

- Usability: Clean, intuitive interface with responsive design for desktop and mobile.
- Performance: Support concurrent attempts by at least 100–200 students.
- Security: Password hashing, prepared statements, role-based access, HTTPS recommended.

- Reliability: Auto-save answers periodically to prevent data loss.
- Maintainability: Modular code structure with clear separation of concerns.
- Scalability: Design should allow horizontal scaling of application and database layers.

3.3 Feasibility Study

Technical Feasibility: The required technologies (HTML, CSS, JavaScript, PHP/Node.js/Python, MySQL) are mature, well-documented, and freely available. Development and hosting can be done with minimal cost.

Economic Feasibility: Open-source stack eliminates licensing costs. Hosting on shared or cloud servers is inexpensive for institutional scale.

Operational Feasibility: The system reduces the workload of teachers and examination cells while providing students with faster feedback, making it highly acceptable.

4. SYSTEM DESIGN

4.1 System Architecture

The Online Quiz Management System follows a classic three-tier architecture consisting of the Presentation (Client) Layer, Application (Business Logic) Layer, and Data Layer. This separation ensures maintainability, security, and scalability.

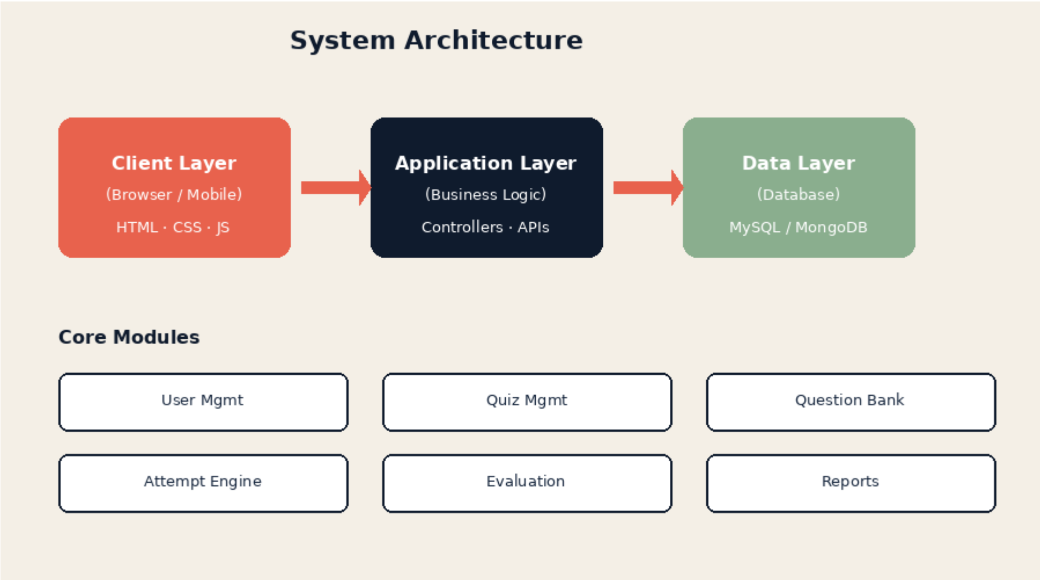

Figure 4.1: System Architecture Diagram

The Client Layer runs in the user's browser and is responsible for rendering the user interface and handling client-side interactions. The Application Layer processes business rules, authentication, quiz logic, and evaluation. The Data Layer stores all persistent information such as users, quizzes, questions, attempts, and results.

4.2 Use Case Diagram

The use case diagram illustrates the primary interactions between the main actors (Student and Teacher) and the system.

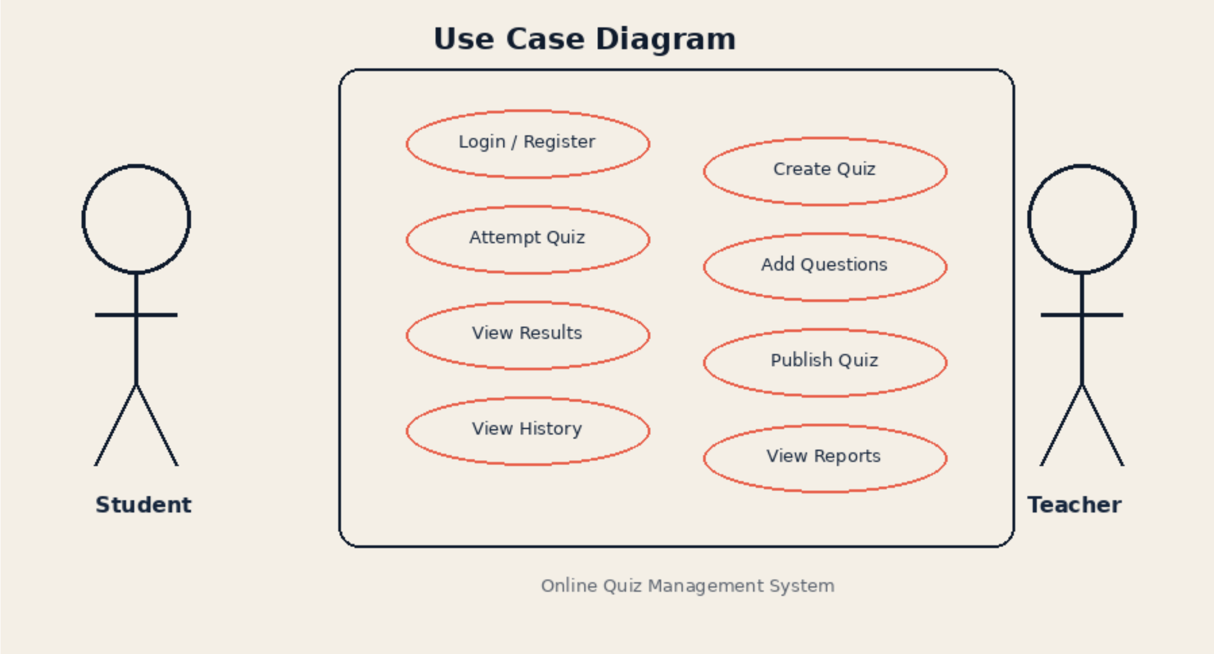

Figure 4.2: Use Case Diagram

4.3 Entity Relationship Diagram

The database is designed around six core entities. The relationships ensure data integrity through primary and foreign keys.

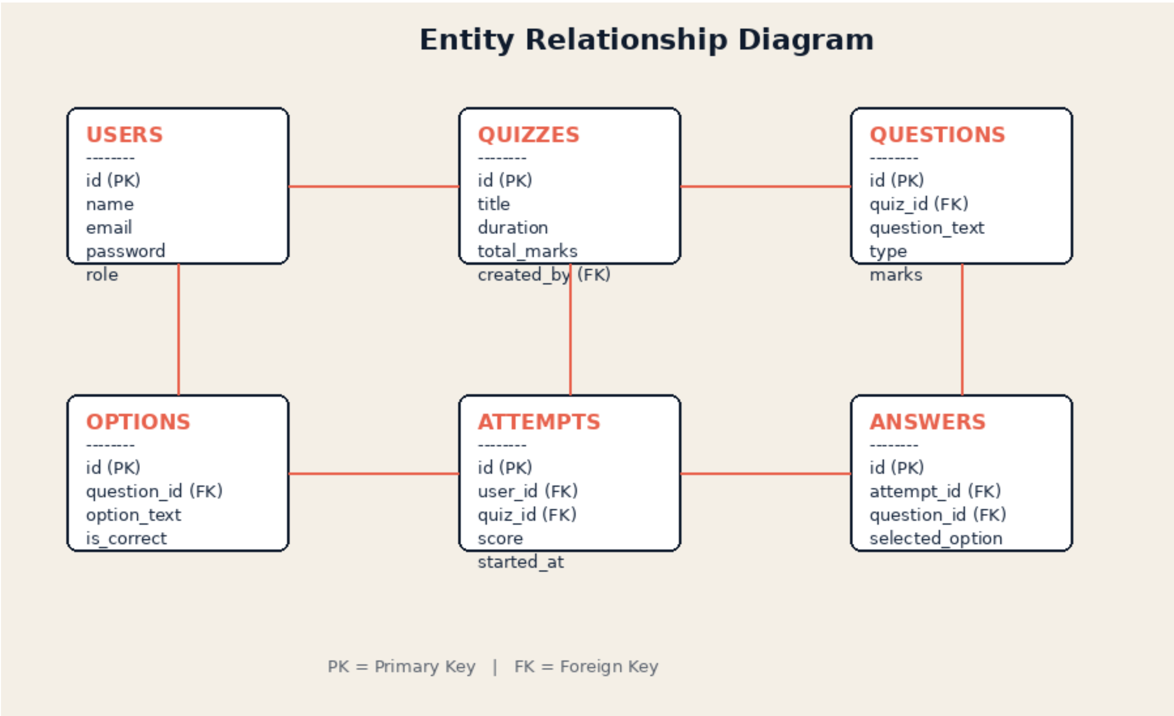

Figure 4.3: Entity Relationship Diagram

4.4 Workflow / Process Flow

The complete lifecycle of a quiz from creation to result publication is shown below.

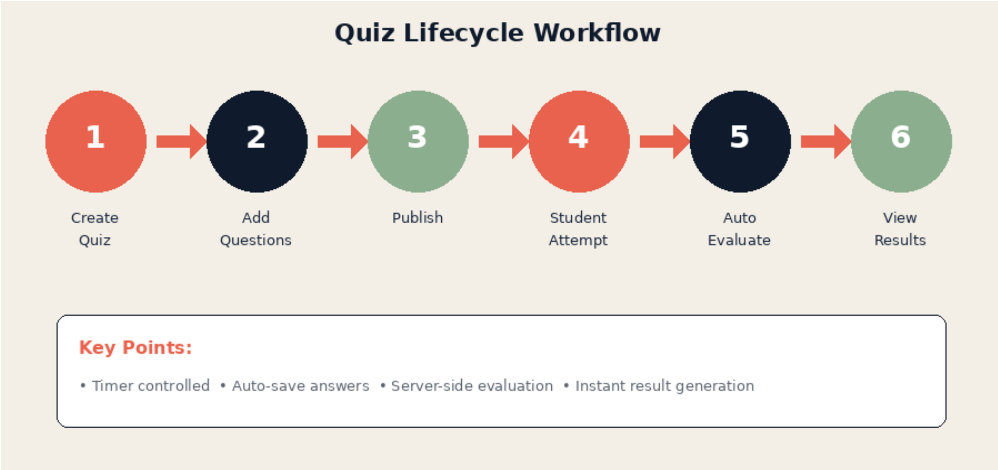

Figure 4.4: Quiz Lifecycle Workflow

5. IMPLEMENTATION

5.1 Technology Stack

The following technologies were used (or recommended) for the development of the system:

Layer	Technology / Tool
Frontend	HTML5, CSS3, Bootstrap / Tailwind CSS, JavaScript
Backend	PHP / Node.js / Python (Django or Flask) / Java
Database	MySQL / MongoDB
Authentication	Session-based or JWT; bcrypt password hashing
Development Tools	VS Code, XAMPP / WAMP, Git, Postman
Server	Apache / Nginx

5.2 Module Description

User Management Module

Handles registration, login, logout, password management, and role assignment. Passwords are stored only in hashed form. Session or token-based authentication ensures that only authorized users access protected resources.

Quiz Management Module

Allows teachers to create quizzes, set parameters (duration, marks, negative marking), and manage the lifecycle of a quiz (draft, published, closed).

Question Bank Module

Supports addition of multiple question types. For MCQs, options are stored separately with a flag indicating the correct answer. Questions can be reused across quizzes.

Attempt Engine

Manages the student attempt process. Starts a timer, records answers, supports auto-save, and forces submission when time expires.

Evaluation & Result Module

Automatically scores objective questions by comparing submitted answers with the correct options stored in the database. Generates score, percentage, and detailed feedback.

Reporting Module

Provides dashboards for teachers and administrators showing class performance, individual student history, and question-wise analysis.

5.3 Database Schema

Key tables and their important columns:

Table	Important Columns
users	id, name, email, password, role, created_at
quizzes	id, title, description, duration, total_marks, created_by
questions	id, quiz_id, question_text, type, marks
options	id, question_id, option_text, is_correct
attempts	id, user_id, quiz_id, score, started_at, submitted_at
answers	id, attempt_id, question_id, selected_option

6. TESTING

Testing was performed at multiple levels to ensure correctness, security, and usability of the system.

6.1 Unit Testing

Individual modules such as login, question addition, score calculation, and timer logic were tested in isolation using test data.

6.2 Integration Testing

End-to-end flows were verified: Teacher creates quiz → Student attempts → System evaluates → Result is displayed.

6.3 System Testing

The complete application was tested under realistic conditions including simultaneous login of multiple users and timed quiz attempts.

6.4 Sample Test Cases

Test Case	Expected Result	Status
Valid login with correct credentials	Dashboard loaded according to role	Pass
Invalid password	Error message displayed	Pass
Create quiz with timer	Quiz saved and timer starts on attempt	Pass
Submit after time expiry	Auto-submit and evaluation occurs	Pass

MCQ evaluation	Correct score calculated	Pass
Unauthorized access to admin page	Access denied / redirect to login	Pass

6.5 Security Testing

Password hashing, SQL injection prevention (prepared statements), and session management were verified. Role-based restrictions were confirmed by attempting to access restricted URLs.

7. RESULTS & DISCUSSION

The Online Quiz Management System was successfully developed and tested. All major functional requirements were met. The system provides a smooth experience for both teachers and students.

Key outcomes observed:

- Teachers can create and publish a complete quiz in a few minutes.
- Students receive instant feedback after submitting a quiz.
- Automatic evaluation eliminates manual checking for objective questions.
- Performance reports help teachers identify weak areas in the class.
- The responsive interface works adequately on mobile browsers.

The modular design allows easy addition of new question types or reporting features in the future. Security measures implemented (hashing, role checks, server-side evaluation) provide a solid foundation for institutional deployment.

Limitations of the current version include the absence of advanced proctoring (webcam monitoring), limited support for complex subjective answer evaluation, and basic analytics. These areas form the basis of the future scope discussed in the next chapter.

8. CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The Online Quiz Management System successfully digitizes the entire examination process — from quiz creation and question management to timed attempts, automatic evaluation, and performance reporting. By providing role-based access and a clean user interface, the system reduces the administrative burden on teachers while offering students a convenient and transparent assessment experience.

This internship project provided valuable hands-on experience in full-stack web development, database design, security best practices, and the complete software development lifecycle. The developed system can serve as a practical tool for educational institutions seeking an affordable and customizable digital assessment solution.

8.2 Future Scope

The following enhancements can be considered for future versions of the system:

1. AI-based question generation from syllabus or uploaded documents.
2. Live proctoring using webcam and AI-based behavior analysis.
3. Native Android and iOS mobile applications with offline support.
4. Integration with popular Learning Management Systems (Moodle, Google Classroom).
5. Advanced analytics and personalized learning recommendations.
6. Multi-language interface and question support.

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APPENDIX

A. Sample Login Credentials (for demonstration)

Admin: admin@example.com / admin123

Teacher: teacher@example.com / teacher123

Student: student@example.com / student123

B. Glossary of Terms

MCQ – Multiple Choice Question

JWT – JSON Web Token

RBAC – Role-Based Access Control

DFD – Data Flow Diagram

ER – Entity Relationship

LMS – Learning Management System

C. Tools & Environment Used

Operating System: Windows / Linux

IDE: Visual Studio Code

Local Server: XAMPP / WAMP / Node.js

Version Control: Git & GitHub

Browser: Google Chrome / Mozilla Firefox

— End of Report —